

Aditya Rohan Sengupta

Email: adityars@ucsc.edu

Website: aditya-sengupta.github.io

Google Scholar: [link](#)

Research interests

Direct imaging, adaptive optics, astrophotonics, control theory, substellar formation and evolution

Appointments and Education

University of California, Santa Cruz

Sep 2022–Jun 2027 expected

PhD Candidate, Astronomy and Astrophysics. **Advisors:** Rebecca Jensen-Clem, Jonathan Fortney

2024 Whitford Prize for highest achievement in research, coursework, and teaching.

2026 Excellence in Mentoring award.

Lawrence Livermore National Laboratory

Jun–Sep 2024

Graduate Intern, Physics Division

University of Cambridge

Oct 2021–Jun 2022

Master of Advanced Study, Part III Applied Mathematics

University of California, Santa Cruz

Jun–Sep 2021

Lamat Research Fellow

University of California, Berkeley

Aug 2017–May 2021

Bachelor of Science, Engineering Physics (Honors) and Engineering Mathematics & Statistics (Honors)

Selected Publications

10. **Sengupta, A.R.**, Chambouleyron, V., Jensen-Clem, R., et al., 2026. “Experimentally-determined performance limits for joint imaging and wavefront sensing with a photonic lantern”. *Proc. SPIE*. (link).
9. **Sengupta, A.R.**, Gerard, B.L., Sanchez, D., et al., 2026. “Laboratory characterization of a multi-photonic lantern optical waveguide using off-axis holography”. *Proc. SPIE*. (link).
8. Gerard, B.L., Sanchez, D., **Sengupta, A.R.**, et al., 2026. “On-sky results from REDWOODS, a platform at Lick/ShaneAO for testing second stage AO technologies”. *Proc. SPIE*. (link).
7. Gerard, B.L., Geringer-Sameth, A., **Sengupta, A.R.**, et al., 2026. “WaveDriver: a Laser Guide Star AO System for HWO”. *Journal of Astronomical Telescopes, Instruments, and Systems*. (link).
6. **Sengupta, A.R.**, Poyneer, L.A., Gerard, B.L., et al., 2026. “Control algorithms for dual-wavefront sensor single-conjugate adaptive optics”. *Publications of the Astronomical Society of the Pacific*. (link).
5. **Sengupta, A.R.**, Diaz, J., DeMartino, M., et al., 2025. “On-sky demonstration of second-stage wavefront control with a photonic lantern”. *The Astronomical Journal*. (link).
4. Cuevas, M., **Sengupta, A.R.**, Chambouleyron, V., et al., 2025. “Exploring the capabilities of astrophotonics for the precise alignment of segmented telescopes”. *Proc. SPIE*. (link).
3. **Sengupta, A.R.**, Diaz, J., Gerard, B.L., et al., 2024. “Photonic lantern wavefront reconstruction in a multi-wavefront sensor single-conjugate adaptive optics system”. *Proc. SPIE*. (link).
2. Chambouleyron, V., **Sengupta, A.R.**, Salama, M., et al., 2024. “Using the Gerchberg–Saxton algorithm to reconstruct non-modulated pyramid wavefront sensor measurements”. *A&A*. (link).
1. **Sengupta, A.R.**, Gerard, B.L., Dillon, D., et al., 2022. “Laboratory demonstration of optimal identification and control of tip-tilt systems”. *Proc. SPIE*. (link).

Selected Talks

Sep 2026	SPLAT Seminar, University of Hawai'i
Jan 2026	Center for Adaptive Optics Retreat, Monterey
Oct 2024	Network of Young Researchers in Instrumentation for Astronomy, San Diego
Sep 2024	Adaptive Optics Group, Lawrence Livermore National Laboratory
Jun 2024	SPIE Astronomical Telescopes & Instrumentation, Yokohama
May 2024	FLASH Seminar, UC Santa Cruz
Dec 2021	WFS Workshop 2021 (Wavefront Sensing in the VLT/ELT Era), Virtual
Jan 2021	American Astronomical Society Meeting, Virtual

Teaching, Mentoring, and Service

Instructor, *Introduction to Astronomical Research and Teaching* Fall 2025, Fall 2026
I developed and taught a course for incoming first-year graduate students, in which I employed active-learning and Universal Design techniques to teach pedagogy and research skills. ([Link to syllabus](#)).

Head Teaching Assistant Sep 2024–present
I assist my department's TAs by connecting them with resources (demonstrations, teaching materials), helping them develop lesson plans, covering absences, and interfacing with faculty when needed.

Graduate Leadership Fellow, Lamat REU 4 summers (2023–present)

- I support 20+ research projects per summer. I serve on the selection committee and help with mentor matching, consult with mentors to make sure their projects are appropriately scoped and meet their mentees' goals, and run check-ins with mentors and mentees throughout the summer.
- I teach a 3-day scientific computing workshop. I run interactive and discussion-based activities to teach computational thinking: they design and play games, 'execute an algorithm' as a class where each person does only one step, and create full computational plans for previous Lamat projects.

Reviewer, *Communications Physics* 2026
Research Mentoring Jun 2023–present

- Maria Cuevas (Columbia University '26, incoming Spärck AI fellow at University of Cambridge)
Project: Laboratory testing of primary-mirror misalignment sensing using a photonic lantern.
- Renata Nobile-Orozco (UCSC '28)
Project: Laboratory demonstration of science image prediction using adaptive optics telemetry.
- UCSC Astronomy 9 (Introduction to Research in Physics and Astrophysics) group
Project: Design and construction of a solar telescope.

Further Teaching Experience Jan 2018–present

- Lecturer for UCSC Astr 215, Research Practicum: *Software Development for Research*
- Lecturer for CfAO Summer School in Adaptive Optics: *Control Theory*
- Lecturer for UCSC Astr 9, Introduction to Research in Physics and Astrophysics: *The Limitations of AI Programming for Astronomy*
- TA for UCSC Astr 118 (Physics of Planetary Systems), Winter 2024; Astr 10 (From the Big Bang to Planet Earth), Spring 2023
- TA for UC Berkeley EECS 126 (Probability and Random Processes) $\times 2$; designed a new computational assignment (Kalman filtering to predict the user's mouse movement) and course notes
- Instructor and Course Designer, Democratic Education at Cal, Spring 2021. *Physics 198: Physics-based High-Performance and Scientific Computing and Technology*. Lectured, designed homework and final project for a 20-student 2-credit pass/fail class
- Drop-In Math Tutor/Study Group Facilitator, UC Berkeley Student Learning Center